

Bird Species Detection Using Deep Learning Models

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- The project focuses on the domain of **computer vision** and **deep learning**, specifically applied to **bird species detection and classification**.
- The goal is to automatically identify bird species from images using advanced neural network models such as the **Vision Transformer (ViT)**, **ResNet Models**, and **other CNN Models**.
- This problem is significant for applications in **biodiversity monitoring**, **wildlife conservation**, and **ecological research**, where manual identification is time-consuming and prone to error.
- The project demonstrates how AI can contribute to **environmental sustainability** and **scientific discovery** through automated visual recognition.

- Identifying bird species manually is challenging and requires expert knowledge, making it difficult for large-scale biodiversity monitoring.
- The growing availability of bird image datasets inspired the idea of applying **deep learning** to automate species recognition.
- A personal interest in **artificial intelligence** and its applications in **environmental conservation** motivated the development of this project.

- To develop an **automated system** capable of accurately detecting and classifying bird species from images.
- To implement and train various models on the **CUB-200-2011 dataset** for high-accuracy recognition.
- To evaluate the model's performances and demonstrate how deep learning can support **wildlife research and conservation efforts**.

- **VGGNet (2014):** Deep CNN with 3×3 filters; improved accuracy through network depth [75.6% Accuracy].
- **Part-based R-CNN (2014):** Combined object and part detection using CNN features [73.9–76.4% Accuracy].
- **RA-CNN (2017):** Attention-based model focusing on key image regions automatically [85.3% Accuracy].
- **Performance:** Accuracy on CUB-200 improved from 73% to 85%.
- **Trend:** Shift from deep features $\rightarrow$ part localization $\rightarrow$ learned attention.

- The workflow begins with collecting and preprocessing the **CUB-200-2011 bird image dataset**, including resizing, normalization, and data augmentation.
- The model is trained using **transfer learning** and **fine-tuning** techniques for improved performance and faster convergence.
- Performance is evaluated using metrics such as **accuracy**, **precision**, and **confusion matrix analysis**.

- The project uses the **CUB-200-2011 (Caltech-UCSD Birds)** dataset, a standard benchmark for fine-grained bird species classification.
- It contains **11,788 images** across **200 bird species**, each annotated with **bounding boxes**, **part locations**, and **class labels**.
- The images are diverse in pose, background, and lighting, providing realistic challenges for model generalization.

Expected Outcome of the Project



- Achievement of a **high classification accuracy** and strong performance metrics such as precision, recall, and F1-score from the selected trained models.
- Development of an **automated bird identification system** that can assist in **biodiversity monitoring**.
- The project's success will be measured by the model's quantitative results and its ability to generalize effectively to unseen images.

- Completed the **literature review** and studied various deep learning models for fine-grained image classification.
- Collected and explored the **CUB-200-2011 dataset**, including data preprocessing and augmentation.
- Implemented the **ResNet18** model architecture and set up the training pipeline. Achieved 55% Accuracy through this model.
- Implemented the **Vision Transformer (ViT-B/16)** model architecture and set up the training pipeline.
- Trained the model for initial epochs and analyzed performance metrics for further optimization.
- **Next Steps:** Fine-tune the model, perform hyperparameter tuning, evaluate on test data, and visualize results through accuracy and confusion matrix plots.

Thank You!

Questions and Discussions